LOVELY PROFESSIONAL UNIVERSITY

Sample Question Set - Unit 5

Course Code: INT396

Course Title: UNSUPERVISED LEARNING

Course Outcome: CO5

Paper: Sample Multiple Choice Questions

Maximum Marks: 40 x 1 = 40

Unit: 5

Unit Title: Association Rule Mining and Anomaly Detection

Unit Mapped RBT: L3 (Apply)

Total Questions: 40

Suggested Duration: 60 Minutes

Answer Grid (to be filled by the student)

Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer
Q1		Q11		Q21		Q31	
Q2		Q12		Q22		Q32	
Q3		Q13		Q23		Q33	
Q4		Q14		Q24		Q34	
Q5		Q15		Q25		Q35	
Q6		Q16		Q26		Q36	
Q7		Q17		Q27		Q37	
Q8		Q18		Q28		Q38	
Q9		Q19		Q29		Q39	
Q10		Q20		Q30		Q40	

Multiple Choice Questions (40 x 1 = 40 Marks)

Q1. Identify the part of an association rule that appears before the arrow.

A) The antecedent
B) The consequent
C) The support
D) The lift

Q2. Name the structure that FP-Growth builds in order to hold the transaction database in compressed form.

A) The proximity matrix
B) The FP-tree
C) The isolation tree
D) The dendrogram

Q3. State what the contamination setting of an Isolation Forest specifies.

A) The number of trees that will be grown
B) The depth to which each tree is grown
C) The expected proportion of anomalies in the data
D) The number of features used per split

Q4. Name the anomaly score that compares the density around a record with the density around its neighbours.

A) The silhouette coefficient
B) The Davies-Bouldin Index
C) The confidence of a rule
D) The Local Outlier Factor

Q5. Identify the value of lift at which two items occur together exactly as often as chance would predict.

A) 0
B) 1
C) 0.5
D) Infinity

Q6. State the number of passes over the transaction database that FP-Growth requires.

A) One
B) One for each item
C) One for each itemset size
D) Two

Q7. Explain why an Isolation Forest separates an anomalous record after very few random splits.

A) It lies in a sparse region and is easily cut off
B) It carries a label marking it as anomalous
C) It is always placed at the root of every tree
D) It has the largest value of every feature

Q8. Distinguish an itemset from an association rule.

A) An itemset is directed, a rule is an unordered set
B) Both of the two are unordered sets of items

C) An itemset is a set, a rule is a directed implication
D) Both of the two are directed implications

Q9. Describe what the conviction of a rule measures.

A) How many transactions contain the antecedent
B) How many items appear in the consequent
C) How often the rule would be wrong if the items were independent
D) How long the algorithm took to find the rule

Q10. Interpret a lift of 0.6 reported for an association rule.

A) The two items occur together less often than by chance
B) The two items occur together more often than chance
C) The rule holds in 60 per cent of all transactions
D) The rule was found in 60 per cent of the passes

Q11. Predict the effect on an Apriori run of lowering the minimum support threshold.

A) Far fewer frequent itemsets and a faster run
B) The same itemsets, found in fewer passes
C) No frequent itemsets can then be found
D) Far more frequent itemsets and a much slower run

Q12. Recognize the application described: a system monitors network traffic and raises an alert on a pattern of connections unlike anything it has recorded before.

A) Association rule mining for cross-selling
B) Anomaly detection in cybersecurity
C) Market basket analysis of purchases
D) Hierarchical clustering of the traffic

Q13. Explain why the Local Outlier Factor can find an outlier that a single global distance threshold misses.

A) It uses the class labels of the records
B) It examines every pair of records in the table
C) It requires the number of outliers in advance
D) It judges a record against its own neighbourhood's density

Q14. Describe what novelty detection assumes about the data it is trained on.

A) That it is fully labelled by an expert
B) That it contains at least half anomalies
C) That it contains no anomalies at all
D) That every feature has been standardised

Q15. Explain why the confidence of a rule is misleading when the consequent is a very common item.

A) Confidence cannot be computed for common items
B) A popular item follows almost any antecedent
C) Confidence falls as the consequent becomes common
D) Confidence is only defined for rare consequents

Q16. Distinguish the support of a rule from its confidence.

A) Support is a share of all transactions, confidence of the antecedent's
B) Confidence is a share of all transactions, support of the antecedent's
C) Both of the two are shares of all the transactions
D) Both of the two are shares of the antecedent's transactions

Q17. Predict what happens to the number of candidate itemsets as the number of distinct items in the catalogue grows.

A) It grows in direct proportion to the items
B) It grows exponentially with the number of items
C) It stays fixed whatever the catalogue holds
D) It falls as more items are added to the catalogue

Q18. Interpret a Local Outlier Factor value close to 1 reported for a record.

A) Its density matches that of its neighbours
B) It is a clear outlier in the dataset
C) It lies at the exact centre of the data
D) It is a duplicate of another record

Q19. Solve the following: a database holds 800 transactions and the itemset of milk with bread appears in 120 of them. Determine its support.

A) 0.12
B) 0.20
C) 1.50
D) 0.15

Q20. Apply the definition of confidence to the rule linking nappies to wipes, given that nappies appear in 400 transactions and nappies with wipes in 260.

A) 0.26
B) 0.40
C) 0.65
D) 1.54

Q21. Use the following to determine the lift of a rule: the support of the rule is 0.10, of the antecedent 0.25 and of the consequent 0.40.

A) 1.0
B) 0.4
C) 2.5
D) 0.1

Q22. Compute the number of non-empty itemsets that can be formed from a catalogue of 5 distinct items.

A) 25
B) 32
C) 31
D) 10

Q23. Solve the following: determine the number of association rules that can be formed from the frequent itemset holding A, B and C.

A) 3
B) 6
C) 8
D) 9

Q24. Apply the Apriori pruning rule to a candidate three-itemset one of whose two-item subsets was found to be infrequent.

A) It is counted in the next database pass
B) It is promoted to the next level directly
C) It is merged with a frequent itemset
D) It is discarded without being counted

Q25. Compute the confidence of a rule for which the support of the combined itemset is 0.12 and the support of the antecedent is 0.30.

A) 0.036
B) 2.5
C) 0.4
D) 0.12

Q26. Use the following to determine whether items A and B are independent: A has support 0.20, B has support 0.50 and the rule has support 0.10.

A) They are positively associated, the lift exceeding 1
B) They are negatively associated, the lift below 1
C) Independence cannot be judged from these figures
D) They are independent, since the lift is exactly 1

Q27. Examine a rule with confidence 0.90 and lift 1.05, and determine what should be concluded.

A) The consequent is very common and the rule adds little
B) The rule is among the strongest that can be found
C) The confidence must have been computed wrongly
D) The antecedent almost never appears in the data

Q28. Apply the Isolation Forest idea to two records, one inside a dense cluster and one far from everything, and determine which gets the shorter average path length.

A) The record inside the dense cluster
B) The record far from everything
C) The two get identical path lengths
D) The path length does not depend on position

Q29. Compare Apriori with FP-Growth on the number of passes each makes over the transaction database.

A) FP-Growth makes many passes, Apriori makes two
B) Both of the two make exactly two passes
C) Both of the two make one pass per itemset
D) Apriori makes many passes, FP-Growth makes two

Q30. Use a contamination setting of 0.01 on a dataset of 50,000 records to determine how many records will be flagged.

A) 50
B) 500
C) 5,000
D) 100

Q31. Solve the following: determine the lift when the rule support is 0.18, the antecedent support 0.30 and the consequent support 0.30.

A) 0.6
B) 1.0
C) 2.0
D) 0.09

Q32. Apply the Local Outlier Factor and determine what a reported value of 2.5 indicates about a record.

A) It sits in a region far sparser than its neighbours
B) It sits in a region far denser than its neighbours
C) It is exactly as dense as its neighbours
D) It is a duplicate of one of its neighbours

Q33. Examine a market basket result in which every high-confidence rule has the same popular staple as its consequent, and determine what has happened.

A) Confidence has been inflated by that item's base rate
B) The minimum support was set far too low
C) The transactions were recorded incorrectly
D) The algorithm failed to prune its candidates

Q34. Use the level-wise structure of Apriori to determine which candidates it examines first.

A) The largest itemsets, before any smaller ones
B) The itemsets holding the most popular item
C) The itemsets appearing in the first transaction
D) The one-item sets, before any larger ones

Q35. Compute the support count corresponding to a minimum support of 0.04 on a database of 2,500 transactions.

A) 40
B) 100
C) 250
D) 625

Q36. Apply the anti-monotone property to determine the support of an itemset relative to the support of any of its subsets.

A) It is never smaller than that of any subset
B) It is always exactly equal to that of a subset
C) It is never greater than that of any subset
D) It bears no relation to that of a subset

Q37. Compare anomaly detection with novelty detection on the data each one is trained upon.

A) Novelty detection trains on contaminated data, anomaly on clean
B) Both of the two train on data known to be clean
C) Anomaly detection trains on contaminated data, novelty on clean
D) Both of the two train on data known to be contaminated

Q38. Examine the effect on an anomaly detector of lowering the threshold on its anomaly score.

A) Fewer records are flagged and false positives rise
B) More records are flagged and false positives rise
C) More records are flagged and false positives fall
D) The number of flagged records does not change

Q39. Use the properties of Isolation Forest and Local Outlier Factor to select a method for scoring ten million records quickly.

A) Local Outlier Factor, which needs no neighbours
B) Local Outlier Factor, whose cost is always lower
C) Neither, since both require the full distance matrix
D) Isolation Forest, whose cost grows roughly linearly

Q40. Demonstrate how the output of a market basket analysis is used inside a store.

A) Associated items are placed together or offered as a bundle
B) Customers are ranked by their total spending
C) Products with low support are removed at once
D) Every transaction is checked for possible fraud

----- End of Sample Question Set -----